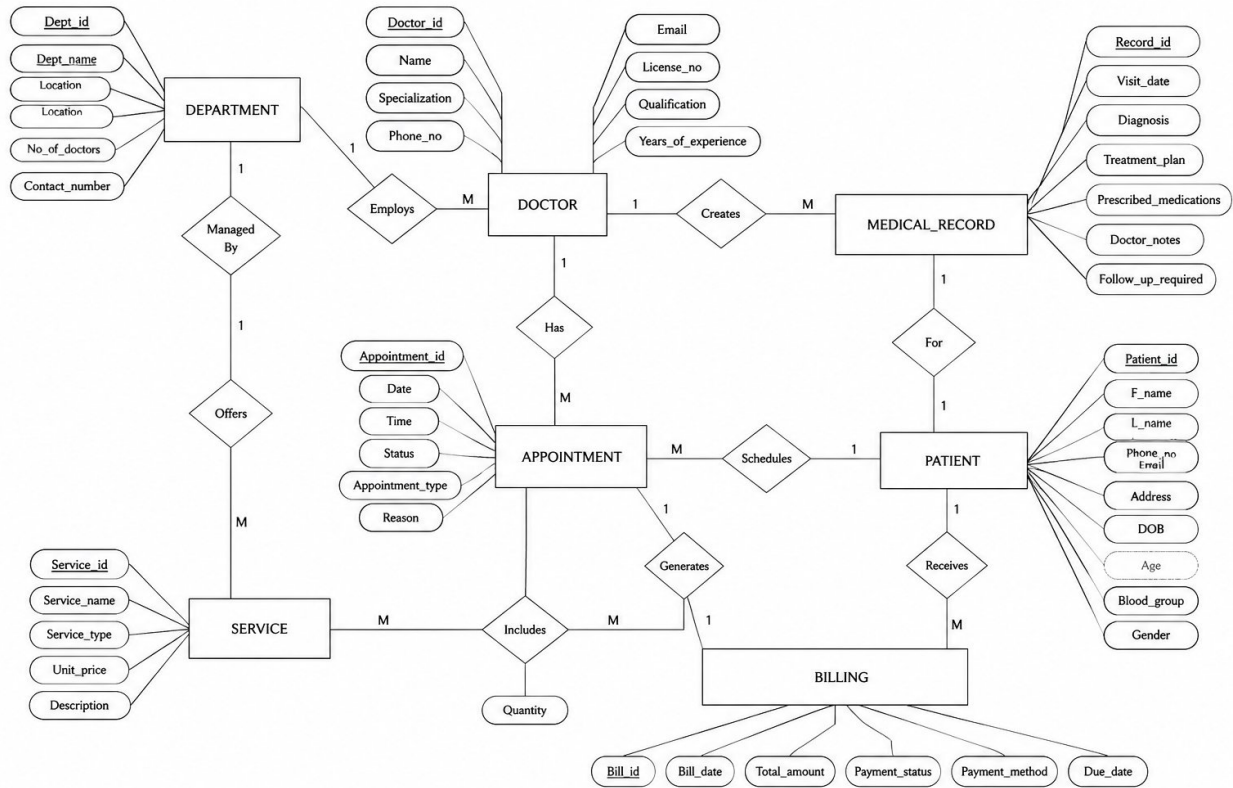


Hospital Management System

Database Design & Implementation

ERD · Mapping · SQL Schema



Section 1: ERD — Entities & Relationships

The system contains 7 main entities and 1 associative entity.

PATIENT

Attribute	Type	Constraint	Note
Patient_id	INT	PRIMARY KEY	Unique identifier
F_name	VARCHAR(75)	NOT NULL	First name
L_name	VARCHAR(75)	NOT NULL	Last name
Phone_no	VARCHAR(20)	NOT NULL	
Email	VARCHAR(100)	UNIQUE	
Address	TEXT	—	
DOB	DATE	NOT NULL	
Age	—	DERIVED	Calculated from DOB, not stored
Blood_group	VARCHAR(5)	—	e.g. A+, O-
Gender	VARCHAR(10)	—	

- 1:M with APPOINTMENT (Schedules)
- 1:M with MEDICAL_RECORD (Has)
- 1:M with BILLING (Receives)

DOCTOR

Attribute	Type	Constraint	Note
Doctor_id	INT	PRIMARY KEY	
Name	VARCHAR(150)	NOT NULL	
Specialization	VARCHAR(100)	NOT NULL	
Phone_no	VARCHAR(20)	NOT NULL	
Email	VARCHAR(100)	UNIQUE	
License_no	VARCHAR(50)	UNIQUE NOT NULL	Medical license
Qualification	VARCHAR(100)	NOT NULL	e.g. MBBS, MD
Years_of_experience	INT	DEFAULT 0	
Dept_id	INT	FOREIGN KEY	References DEPARTMENT
Supervisor_id	INT	FOREIGN KEY	Self-ref: supervising doctor

- M:1 with DEPARTMENT (Works in)
- 1:M with APPOINTMENT
- 1:M with MEDICAL_RECORD (Creates)
- Recursive: supervised by another DOCTOR

DEPARTMENT

Attribute	Type	Constraint	Note
Dept_id	INT	PRIMARY KEY	
Dept_name	VARCHAR(100)	NOT NULL	
Location	VARCHAR(150)	—	
No_of_doctors	INT	DEFAULT 0	
Contact_number	VARCHAR(20)	—	
Head_Doctor_id	INT	FOREIGN KEY	References DOCTOR

- 1:M with DOCTOR (Employs)
- 1:1 with DOCTOR (Managed By)
- 1:M with SERVICE (Offers)

APPOINTMENT

Attribute	Type	Constraint	Note
Appointment_id	INT	PRIMARY KEY	
Date	DATE	NOT NULL	
Time	TIME	NOT NULL	
Status	VARCHAR(20)	CHECK	Scheduled / Completed / Cancelled
Appointment_type	VARCHAR(20)	CHECK	Consultation / Follow-up / Emergency
Reason	TEXT	—	
Patient_id	INT	FOREIGN KEY	References PATIENT
Doctor_id	INT	FOREIGN KEY	References DOCTOR

- M:M with SERVICE via APPOINTMENT_SERVICE
- 1:1 with MEDICAL_RECORD (Results In)
- 1:1 with BILLING (Generates)

SERVICE

Attribute	Type	Constraint	Note
Service_id	INT	PRIMARY KEY	
Service_name	VARCHAR(150)	NOT NULL	
Service_type	VARCHAR(50)	CHECK	Consultation/Lab Test/X-Ray/Surgery/Treatment
Unit_price	DECIMAL(10,2)	NOT NULL	
Description	TEXT	—	
Dept_id	INT	FOREIGN KEY	References DEPARTMENT

- M:1 with DEPARTMENT
- M:M with APPOINTMENT via APPOINTMENT_SERVICE

APPOINTMENT_SERVICE (Associative — resolves M:M)

Attribute	Type	Constraint	Note
Appointment_id	INT	PK / FK	References APPOINTMENT
Service_id	INT	PK / FK	References SERVICE
Quantity	INT	NOT NULL DEFAULT 1	Number of times service rendered

MEDICAL_RECORD

Attribute	Type	Constraint	Note
Record_id	INT	PRIMARY KEY	
Visit_date	DATE	NOT NULL	
Diagnosis	TEXT	—	
Treatment_plan	TEXT	—	
Prescribed_medications	TEXT	—	
Doctor_notes	TEXT	—	
Follow_up_required	BOOLEAN	DEFAULT FALSE	
Patient_id	INT	FOREIGN KEY	References PATIENT
Doctor_id	INT	FOREIGN KEY	References DOCTOR
Appointment_id	INT	UNIQUE FK	References APPOINTMENT

BILLING

Attribute	Type	Constraint	Note
Bill_id	INT	PRIMARY KEY	
Bill_date	DATE	NOT NULL	
Total_amount	DECIMAL(10,2)	NOT NULL	Sum of Unit_price x Quantity per service
Payment_status	VARCHAR(20)	CHECK	Paid / Pending / Partial
Payment_method	VARCHAR(20)	CHECK	Cash / Card / Insurance
Due_date	DATE	—	
Patient_id	INT	FOREIGN KEY	References PATIENT
Appointment_id	INT	UNIQUE FK	References APPOINTMENT

Section 2: Mapping — ERD to Relational Schema

Rule	Description
Strong Entity	Each entity becomes a table. Simple attributes become columns. PK of entity = PK of table.
Derived Attribute	Age is NOT stored. Calculated: TIMESTAMPDIFF(YEAR, DOB, CURDATE()).
1:M Relationship	FK placed on the 'many' side. e.g. Doctor.Dept_id → DEPARTMENT.
M:M Relationship	New table APPOINTMENT_SERVICE created with composite PK (Appointment_id, Service_id) + Quantity.
1:1 Relationship	DEPARTMENT.Head_Doctor_id FK → DOCTOR. BILLING and MEDICAL_RECORD have Appointment_id as UNIQUE FK.
Recursive Relationship	DOCTOR.Supervisor_id FK → DOCTOR(Doctor_id). Models supervision within the same table.

Final Relational Schema

Table	Columns
DEPARTMENT	Dept_id (PK), Dept_name, Location, No_of_doctors, Contact_number, Head_Doctor_id (FK)
DOCTOR	Doctor_id (PK), Name, Specialization, Phone_no, Email, License_no, Qualification, Years_of_experience, Dept_id (FK), Supervisor_id (FK)
PATIENT	Patient_id (PK), F_name, L_name, Phone_no, Email, Address, DOB, Blood_group, Gender
APPOINTMENT	Appointment_id (PK), Date, Time, Status, Appointment_type, Reason, Patient_id (FK), Doctor_id (FK)
SERVICE	Service_id (PK), Service_name, Service_type, Unit_price, Description, Dept_id (FK)

APPOINTMENT_SERVICE	Appointment_id (PK,FK), Service_id (PK,FK), Quantity
MEDICAL_RECORD	Record_id (PK), Visit_date, Diagnosis, Treatment_plan, Prescribed_medications, Doctor_notes, Follow_up_required, Patient_id (FK), Doctor_id (FK), Appointment_id (FK,UNIQUE)
BILLING	Bill_id (PK), Bill_date, Total_amount, Payment_status, Payment_method, Due_date, Patient_id (FK), Appointment_id (FK,UNIQUE)

Section 3: SQL Implementation

Database

```
CREATE DATABASE IF NOT EXISTS HospitalDB;
USE HospitalDB;
```

DEPARTMENT

```
CREATE TABLE DEPARTMENT (
  Dept_id INT PRIMARY KEY AUTO_INCREMENT,
  Dept_name VARCHAR(100) NOT NULL,
  Location VARCHAR(150),
  No_of_doctors INT DEFAULT 0,
  Contact_number VARCHAR(20),
  Head_Doctor_id INT,
  FOREIGN KEY (Head_Doctor_id) REFERENCES DOCTOR(Doctor_id) ON DELETE SET NULL
);
```

DOCTOR

```
CREATE TABLE DOCTOR (
  Doctor_id INT PRIMARY KEY AUTO_INCREMENT,
  Name VARCHAR(150) NOT NULL,
  Specialization VARCHAR(100) NOT NULL,
  Phone_no VARCHAR(20) NOT NULL,
  Email VARCHAR(100) UNIQUE,
  License_no VARCHAR(50) UNIQUE NOT NULL,
  Qualification VARCHAR(100) NOT NULL,
  Years_of_experience INT DEFAULT 0,
  Dept_id INT,
  Supervisor_id INT,
  FOREIGN KEY (Dept_id) REFERENCES DEPARTMENT(Dept_id) ON DELETE SET NULL,
  FOREIGN KEY (Supervisor_id) REFERENCES DOCTOR(Doctor_id) ON DELETE SET NULL
);
```

PATIENT

```
CREATE TABLE PATIENT (
  Patient_id INT PRIMARY KEY AUTO_INCREMENT,
  F_name VARCHAR(75) NOT NULL,
  L_name VARCHAR(75) NOT NULL,
  Phone_no VARCHAR(20) NOT NULL,
  Email VARCHAR(100) UNIQUE,
  Address TEXT,
  DOB DATE NOT NULL,
  -- Age is derived: TIMESTAMPDIFF(YEAR, DOB, CURDATE())
  Blood_group VARCHAR(5),
  Gender VARCHAR(10)
);
```

APPOINTMENT

```
CREATE TABLE APPOINTMENT (  
  Appointment_id INT PRIMARY KEY AUTO_INCREMENT,  
  Date DATE NOT NULL,  
  Time TIME NOT NULL,  
  Status VARCHAR(20) NOT NULL  
  CHECK (Status IN ('Scheduled', 'Completed', 'Cancelled')),  
  Appointment_type VARCHAR(20) NOT NULL  
  CHECK (Appointment_type IN ('Consultation', 'Follow-up', 'Emergency')),  
  Reason TEXT,  
  Patient_id INT NOT NULL,  
  Doctor_id INT NOT NULL,  
  FOREIGN KEY (Patient_id) REFERENCES PATIENT(Patient_id) ON DELETE CASCADE,  
  FOREIGN KEY (Doctor_id) REFERENCES DOCTOR(Doctor_id) ON DELETE CASCADE  
);
```

SERVICE

```
CREATE TABLE SERVICE (  
  Service_id INT PRIMARY KEY AUTO_INCREMENT,  
  Service_name VARCHAR(150) NOT NULL,  
  Service_type VARCHAR(50) NOT NULL  
  CHECK (Service_type IN ('Consultation', 'Lab Test', 'X-Ray', 'Surgery', 'Treatment')),  
  Unit_price DECIMAL(10,2) NOT NULL,  
  Description TEXT,  
  Dept_id INT,  
  FOREIGN KEY (Dept_id) REFERENCES DEPARTMENT(Dept_id) ON DELETE SET NULL  
);
```

APPOINTMENT_SERVICE

```
CREATE TABLE APPOINTMENT_SERVICE (  
  Appointment_id INT NOT NULL,  
  Service_id INT NOT NULL,  
  Quantity INT NOT NULL DEFAULT 1,  
  PRIMARY KEY (Appointment_id, Service_id),  
  FOREIGN KEY (Appointment_id) REFERENCES APPOINTMENT(Appointment_id) ON DELETE CASCADE,  
  FOREIGN KEY (Service_id) REFERENCES SERVICE(Service_id) ON DELETE CASCADE  
);
```

MEDICAL_RECORD

```
CREATE TABLE MEDICAL_RECORD (  
  Record_id INT PRIMARY KEY AUTO_INCREMENT,  
  Visit_date DATE NOT NULL,  
  Diagnosis TEXT,  
  Treatment_plan TEXT,  
  Prescribed_medications TEXT,  
  Doctor_notes TEXT,  
  Follow_up_required BOOLEAN DEFAULT FALSE,  
  Patient_id INT NOT NULL,  
  Doctor_id INT NOT NULL,  
  Appointment_id INT UNIQUE NOT NULL,  
  FOREIGN KEY (Patient_id) REFERENCES PATIENT(Patient_id) ON DELETE CASCADE,  
  FOREIGN KEY (Doctor_id) REFERENCES DOCTOR(Doctor_id) ON DELETE CASCADE,  
  FOREIGN KEY (Appointment_id) REFERENCES APPOINTMENT(Appointment_id) ON DELETE CASCADE  
);
```

BILLING

```
CREATE TABLE BILLING (  
  Bill_id INT PRIMARY KEY AUTO_INCREMENT,  
  Bill_date DATE NOT NULL,  
  Total_amount DECIMAL(10,2) NOT NULL,  
  Payment_status VARCHAR(20) NOT NULL  
  CHECK (Payment_status IN ('Paid','Pending','Partial')),  
  Payment_method VARCHAR(20)  
  CHECK (Payment_method IN ('Cash','Card','Insurance')),  
  Due_date DATE,  
  Patient_id INT NOT NULL,  
  Appointment_id INT UNIQUE NOT NULL,  
  FOREIGN KEY (Patient_id) REFERENCES PATIENT(Patient_id) ON DELETE CASCADE,  
  FOREIGN KEY (Appointment_id) REFERENCES APPOINTMENT(Appointment_id) ON DELETE CASCADE  
);
```

View: Derived Age

```
CREATE VIEW PatientWithAge AS  
SELECT *, TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age  
FROM PATIENT;
```

Section 4: Relationship Summary

Relationship	Entities	Cardinality	Resolved By
Schedules	PATIENT — APPOINTMENT	1 : M	Patient_id FK in APPOINTMENT
Has	DOCTOR — APPOINTMENT	1 : M	Doctor_id FK in APPOINTMENT
Employs	DEPARTMENT — DOCTOR	1 : M	Dept_id FK in DOCTOR
Managed By	DEPARTMENT — DOCTOR	1 : 1	Head_Doctor_id FK in DEPARTMENT
Offers	DEPARTMENT — SERVICE	1 : M	Dept_id FK in SERVICE
Includes	APPOINTMENT — SERVICE	M : M	APPOINTMENT_SERVICE + Quantity
Creates	DOCTOR — MEDICAL_RECORD	1 : M	Doctor_id FK in MEDICAL_RECORD
For	PATIENT — MEDICAL_RECORD	1 : M	Patient_id FK in MEDICAL_RECORD
Results In	APPOINTMENT — MEDICAL_RECORD	1 : 1	Appointment_id UNIQUE FK
Generates	APPOINTMENT — BILLING	1 : 1	Appointment_id UNIQUE FK
Receives	PATIENT — BILLING	1 : M	Patient_id FK in BILLING
Supervises	DOCTOR — DOCTOR	1 : M	Supervisor_id self-FK in DOCTOR